Learning rate selection for Gibbs posteriors

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Introduction

> *Statisticians want numerical measures of the degree to which data support hypotheses.* —Hacking

- Problem setup:
  - unknown quantity of interest, $\Theta \in \mathbb{T}$
  - iid data $Z^n = (Z_1, \ldots, Z_n)$ carrying info about Θ
- Goal: uncertainty quantification about Θ, given $Z^n = z^n$
- Bayesian inference is one option,[2] but there are obstacles:
  - model misspecification
  - model under-specification and marginalization
  - prior choice
  - ...
- Post-Bayes seminar is about overcoming these obstacles

[2] Bayes isn't the only brand of UQ; see, e.g., `arXiv:2211.14567`

Intro, cont.

> *The idea that statistical problems do not have to be solved as one coherent whole is anathema to Bayesians but is liberating for frequentists.* —Wasserman

- Under-specification is under-appreciated
- Θ often isn't a parameter that determine a model
- e.g., quantiles, MCID, ML estimands, ...
- Bayesians need a model, but this introduces
  - risk of model misspecification bias
  - nuisance parameters and a need for marginalization
  - challenges using informative priors
- This is bigger than accommodating misspecification
- Generalized Bayes is liberated Bayes!

Intro, cont.

- Distribution of Z^n is P
- There's a loss function $\ell_\theta(z)$ such that

$$\Theta = \theta(P) = \arg\min_\theta R(\theta), \quad \text{where} \quad \underbrace{R(\theta) = P\ell_\theta}_{\text{risk, expected loss}} .$$

- Gibbs posterior for Θ, given $Z^n = z^n$, is

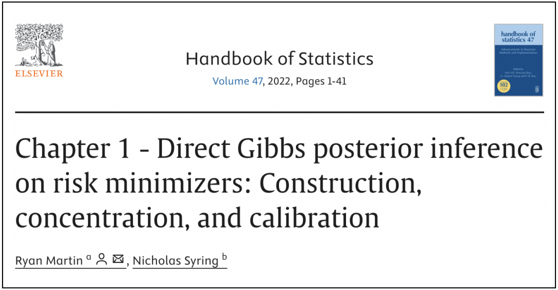
$$\Pi_n^{(\omega)}(d\theta) \propto e^{-\omega n R_n(\theta)} \, \Pi(d\theta)$$

  - Π is a prior distribution for Θ
  - $R_n(\theta) = \mathbb{P}_n \ell_\theta = n^{-1} \sum_{i=1}^n \ell_\theta(z_i)$, empirical risk
- This talk focuses on the *learning rate parameter*, $\omega > 0$

This talk

- background on Gibbs posteriors
    - motivation
    - construction, appearance of learning rate
    - recap of posterior concentration
- learning rate selection methods
- calibration and the GPC algorithm
- illustrations and comparisons
- other things (time permitting):
    - prediction problems
    - anytime-valid inference
- concluding remarks, open problems, etc.

- I don't give comprehensive references in the slides
- Most of what I'll say is in this recent-ish review paper[3]



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Handbook of Statistics

Volume 47, 2022, Pages 1-41

Chapter 1 - Direct Gibbs posterior inference on risk minimizers: Construction, concentration, and calibration

Ryan Martin [a], Nicholas Syring [b]

[3] M. and Syring, `arXiv:2203.09381`

My motivating example

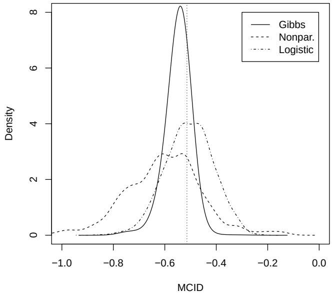
- When do patients feel like a treatment was effective?
- Statistical versus practical/clinical significance...
- Data $Z_i = (X_i, Y_i)$, $i = 1, \ldots, n$
  - $X_i \in \mathbb{R}$ is a diagnostic measure on patient i
  - $Y_i \in \{\pm 1\}$, depending on whether patient i found the treatment to be effective/ineffective
- MCID = "minimum clinically important difference"[4]

$$\Theta = \arg\min_{\theta} \underbrace{P\{Y \neq \mathrm{sign}(X - \theta)\}}_{R(\theta)}$$

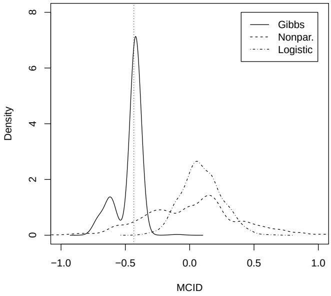
- MCID is the X-scale cutoff at which patients will tend to feel the treatment was effective.

[4] Hedayat et al, `arXiv:1307.3646`

- Compare Gibbs ($\omega = 1$) to two Bayes posteriors[5]
  - logistic regression
  - binary regression with nonparametric link



(a) “Good case”



(b) “Bad case”

[5] Syring and M., `arXiv:1501.01840`

Construction

- Decision-theoretic view:

$$\Pi_n^{(\omega)} = \arg\min_{\Psi} \Big\{ \omega \underbrace{\int nR_n(\theta)\, \Psi(d\theta)}_{\text{chase data } Z^n} + \underbrace{\mathrm{KL}(\Psi, \Pi)}_{\text{stay near prior } \Pi} \Big\}$$

- Solution amounts to using Bayes's rule with a fake likelihood

$$\Pi_n^{(\omega)}(d\theta) \propto \underbrace{e^{-\omega n R_n(\theta)}}_{\text{"likelihood"}} \Pi(d\theta)$$

- ω controls the rate at which info about Θ from the data accumulates relative to info about Θ in the prior

Construction, cont.

- *Learning rate can't be ignored*
- Shape of the loss is real, but scale is often artificial
  - we pick, e.g., squared-error loss for its shape, not because the numerical values exactly represent loss suffered
  - so, $\ell_\theta(z)$ and $\omega\ell_\theta(z)$ are the "same"
- Note: scale ambiguity doesn't affect M-estimators
- Can't estimate ω how we estimate model parameters
  - h-Bayes works when there's a model tying everything together
  - here we specifically don't have a (correct) model
  - so data aren't directly informative about ω

Construction, cont.

- Special/edge case: misspecified models
- If the posited model P_θ has density p_θ, then take the loss function to be $\ell_\theta(z) = -\log p_\theta(z)$
- Then $R(\theta) = \mathsf{KL}(P, P_\theta)$, and Θ is the "KL-best" value
- Wrong likelihood implies scaling ambiguity, so ω still needed
- Gibbs posterior aka *power posterior*
- But don't forget: Gibbs $\neq$ misspecified model Bayes!
  - misspecification on purpose, versus
  - misspecification on accident

Convergence

- Intuition:
  - $\Pi_n^{(\omega)}(d\theta)$ is maximized near where R_n is minimized
  - $R_n(\theta)$ converges to $R(\theta)$ for each θ
  - $R(\theta)$ is minimized at Θ
- So, if $\Pi_n^{(\omega)}$ is concentrating anywhere, then it must be at Θ
- See David Frazier's talk from previous session!
- Better results are possible,[6] but I like the following...

Theorem (M. & Syring, `arXiv:2203.09381`).

Under the standard sufficient conditions for M-estimator consistency, if Π puts enough mass at Θ, then $\Pi_n^{(\omega)} \xrightarrow{\text{P}} \delta_\Theta$ for all $\omega > 0$

[6] Miller, `arXiv:1907.09611`

- More can be said, with some additional regularity
- Specifically, R must be twice differentiable, with $\ddot{R}(\Theta) > 0$
- The *Bernstein–von Mises theorem*[7] says, roughly
  - $\Pi_n^{(\omega)}$ is centered around Θ
  - asymptotic variance $\omega^{-1}\ddot{R}(\Theta)^{-1}$
- Note that the M-estimator asymptotic variance is

$$\underbrace{\ddot{R}(\Theta)^{-1}\bigl(P\dot{\ell}_\Theta\dot{\ell}_\Theta^\top\bigr)\ddot{R}(\Theta)^{-1}}_{\text{sandwich covariance}} \neq \ddot{R}(\Theta)^{-1}$$

- So, Gibbs with $\omega \equiv 1$ could give very bad UQ
- Learning rate might be annoying, but it offers an opportunity to correct for this variance mismatch...

[7] Kleijn & van der Vaart, 2012; M. & Syring, `arXiv:2203.09381`

Learning rate choice

- Data-driven strategies for choosing ω?
- Except that they don't try to "estimate" ω like a model parameter, they're very/completely different
- Non-exhaustive list:

  Grünwald. minimizes cumulative posterior risk
  Holmes & Walker. matches info between data and prior
  Lyddon et al. bootstrap to match info, asymptotic calibration
  Syring & M. bootstrap specifically for calibration

- None is "right," just different priorities/tastes/opinions
- It's my presentation, so I get to share my opinion :)

Calibration

> *[Bayes's formula] does not create real probabilities from hypothetical probabilities.* —Fraser

- Probabilities can be useful even if they aren't "real"
- Most of us are in the business of developing methods and software for broad use, so *reliability* is imperative
- You'd probably only use my method if I can prove to you that it's likely going to give you the right answer
- So, posterior ought to be calibrated in the sense that it tends to assign high/low probability to true/false hypotheses

Calibration, cont.

- Formalize the discussion of calibration...
- $C_\alpha(Z^n;\omega)$ = the $100(1-\alpha)\%$ credible region from $\Pi_n^{(\omega)}$
- If $C_\alpha(Z^n;\omega)$ is also a $100(1-\alpha)\%$ confidence region, then posterior is *calibrated*
- Define the coverage probability function

$$c_\alpha(\omega \mid P) = P\{C_\alpha(Z^n;\omega) \ni \theta(P)\}$$

- Goal is to find ω that (roughly) solves the equation

$$c_\alpha(\omega \mid P) = 1 - \alpha$$

Calibration, cont.

- Does a solution exist?
  - Claim: ω mainly controls the spread of $\Pi_n^{(\omega)}$
    - small ω, $\Pi_n^{(\omega)}$ is close to the prior, relatively diffuse
    - large ω, $\Pi_n^{(\omega)}$ is close to $\delta_{\hat{\theta}}$ ($\approx \delta_\Theta$)
  - Then $c_\alpha(\omega) \to 1$ and $\to 0$ as $\omega \to 0$ and $\to \infty$, resp.
  - $\omega \mapsto c_\alpha(\omega)$ should be continuous (and non-increasing)
  - Therefore, a (unique) solution to $c_\alpha(\eta) = 1 - \alpha$ exists
- More intuition:
  - limiting Gaussian posterior from the BvM
  - "posterior variance" is strictly decreasing in ω
  - find ω to achieve (conservative) calibration

Calibration, cont.

- How to solve $c_\alpha(\omega \mid P) = 1 - \alpha$?
- *If we knew* P, then we could simulate new data Z^n and
  - get a Monte Carlo estimate of $c_\alpha(\omega \mid P)$
  - use Robbins–Monro to solve the equation

Oracle GPC algorithm

Initialize ω_0, and for $t = 0, 1, 2, \ldots$ do the following:

- Simulate data sets Z_b^n, $b = 1, \ldots, B$, from P
- Approximate the $\Pi_n^{(\omega_t)}$-coverage probability by

$$\hat{c}_\alpha(\omega_t \mid P) = \frac{1}{B} \sum_{b=1}^{B} 1\{C_\alpha(Z_b^n; \omega_t) \ni \theta(P)\}$$

- Update $\omega_{t+1} = \omega_t + k_t\{\hat{c}_\alpha(\omega_t \mid P) - (1 - \alpha)\}$

Calibration, cont.

- Of course, we don't know P...
- To get a practical algorithm, replace P with $\mathbb{P}_n$
- This amounts to a bootstrap approx of the coverage prob
- $\theta(P)$ is replaced by $\theta(\mathbb{P}_n)$, emp risk min/MLE

GPC algorithm (Syring and M., `arXiv:1509.00922`)

Initialize ω_0, and for $t = 0, 1, 2, \ldots$ do the following:

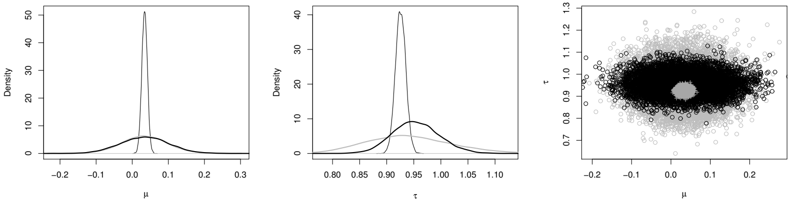
- Simulate data sets Z_b^n, $b = 1, \ldots, B$, from $\mathbb{P}_n$
- Get $\hat{c}_\alpha(\omega_t \mid \mathbb{P}_n) = B^{-1} \sum_{b=1}^{B} 1\{C_\alpha(Z_b^n; \omega_t) \ni \theta(\mathbb{P}_n)\}$
- Update $\omega_{t+1} = \eta_t + k_t\{\hat{c}_\alpha(\omega_t \mid \mathbb{P}_n) - (1 - \alpha)\}$

Calibration, cont.

- GPC isn't cheap — repeated posterior computations
- However:
  - computations at common ω can be parallelized
  - no benefit to large B or strong convergence criterion
- In my experience, with the following settings, computations are manageable and empirical results are good
  - B roughly 200–300
  - cap the number of ω updates at 10–20
- My computational skills are limited, surely someone here can develop a more efficient implementation

Illustrations

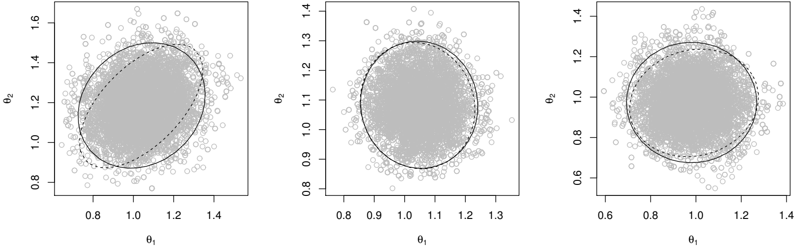
- Simple Gaussian process model with two parameters (μ, τ).
- Compare several posteriors:
  - full Bayes posterior
  - composite Bayes posterior
  - GPC-calibrated version of the composite posterior



(c) μ marginal (d) τ marginal (e) (μ, τ) joint

Illustrations, cont.

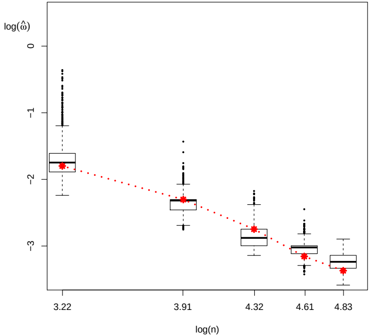
- A few cases with the multivariate median[8]
  - Gibbs posterior credible region (solid)
  - M-estimator confidence region (dashed)
- Left is normal data, right are variations on Laplace



[8] Bhattacharya and M., `arXiv:2002.01052`

Illustrations, cont.

- The oracle GPC algorithm returns $\omega_n^\star$
- Let's compare with $\hat{\omega}_n$ based on the regular GPC
- Plot of $\omega_n^\star$ and boxplot of $\hat{\omega}_n$ versus n (on log scale)[9]



[9] Wang and M. (arXiv:1906.08296)

Illustrations, cont.

- Two-dim quantile regression example
- 95% intervals based on several methods:

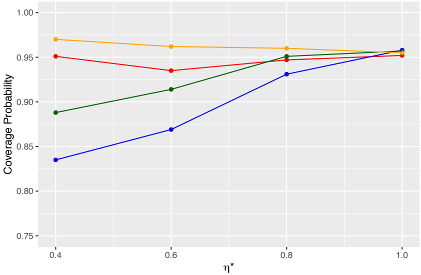
BEL.s Bayesian empirical likelihood
Normal asymptotic normality
GPC Gibbs with GPC

		Coverage Prob $\times 100$			Avg Length $\times 100$		
n		BEL.s	Normal	GPC	BEL.s	Normal	GPC
100	θ_0	97	95	95	106	100	91
	θ_1	98	98	95	58	55	47
400	θ_0	95	95	95	50	50	46
	θ_1	97	97	95	26	25	23
1600	θ_0	96	96	95	25	25	23
	θ_1	96	96	95	13	12	11

Comparisons

- Consider a normal model, misspecified only in the variance
- Oracle $\omega^\star$: ratio of posited variance to true variance
- Coverage probability comparisons for four different methods:

Grünwald
Holmes & Walker
Lyddon et al
GPC



Comparisons, cont.

- Linear regression model that assumes the usual Gaussian, homoscedastic errors
- Misspecification: non-Gaussian or heteroscedastic errors
- Snippets of two (extreme) comparisons[10]

	Coverage prob of 95% intervals	
Method	Non-Gaussian	Heteroscedastic
Grünwald	0.97	0.87
Holmes & Walker	0.96	0.80
Lyddon et al	0.81	0.38
Syring & M.	0.97	0.98

[10] Wu and M., `arXiv:2012.11349`

Comparisons, cont.

Bayesian Analysis (2023) **18**, Number 1, pp. 105–132

A Comparison of Learning Rate Selection Methods in Generalized Bayesian Inference*

Pei-Shien Wu[†] and Ryan Martin[‡]

- These and other comparisons in a recent-ish review
- *Take-away message:* Only GPC calibrates over a range of sample sizes and degrees of misspecification
- Not a criticism — other methods aren't designed for this
- Further comparisons considering other metrics?

Extras

- Time permitting:
  - prediction
  - anytime-valid inference

Extras: prediction

- Suppose Θ represents the "true" parameter in a possibly misspecified model
- Posterior $\Pi_n^{(\omega)}$ for Θ can be pushed forward to a predictive distribution for a new observation Z_{n+1}
- But if we thought the model might be misspecified, would we just average the likelihood as usual to get a predictive?
- We could instead tune a "learning rate" specifically for calibration in prediction!
- Proposed predictive distribution:

$$f_n^{(\eta)}(z) = \int \underbrace{p_\theta^\eta(z)}_{\text{power likelihood}} \underbrace{\Pi_n(d\theta)}_{\text{ordinary posterior}}$$

Extras: prediction, cont.

GPrC algorithm (Wu and M., `arXiv:2107.01688`)

Initialize η_0, and for $t = 0, 1, 2, \ldots$ do the following:

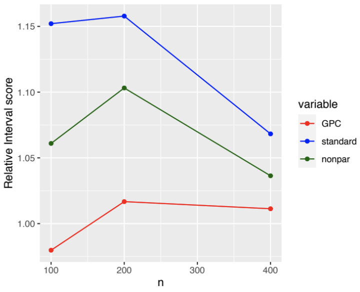
- Simulate data sets Z_b^n, $b = 1, \ldots, B$, from $\mathbb{P}_n$
- Approximate the prediction coverage prob

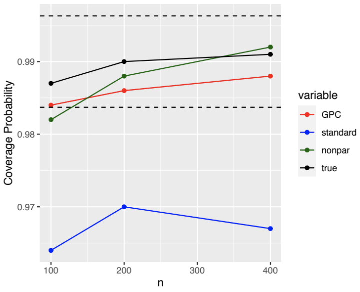
$$\hat{c}_\alpha(\eta_t) = \frac{1}{nB} \sum_{b=1}^{B} \sum_{i=1}^{n} 1\{ \underbrace{Q_\alpha(Z_b^n; \eta_t)}_{(1-\alpha)\text{-quantile of } f_n^{(\eta_t)}} \geq Z_i \}$$

- Update $\eta_{t+1} = \eta_t + k_t\{\hat{c}_\alpha(\eta_t) - (1-\alpha)\}$

Extras: prediction, cont.

- log-normal model, GEV truth
- $\alpha = 0.01$, extreme tail
- Comparison:[11]
  - True, GPrC, Naive, DP mixture
  - interval score (left) and coverage prob (right)





[11] Wu and M., `arXiv:2107.01688`

Extras: anytime-valid inference

- Lots of recent interest in *anytime-valid inference*[12]
- Goal is the construction of tests and confidence sets that are valid under optional stopping and/or continuation
- Theory is based on *e-values/e-processes*
- In model-based anytime-valid inference, the optimal e-values often turn out to be Bayes factors
- *Conjecture+.* For inference on risk minimizers, the optimal e-values often turn out to be *Gibbs factors*
- Learning rates enter the picture...

[12] Grünwald, de Heide, Koolen, Ramdas, Shafer, Vovk, Wang, and others

Extras: anytime-valid inference, cont.

Generalized Universal Inference on Risk Minimizers

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- *Generalized universal inference*[13] also has a Gibbs flavor, working with a "likelihood ratio" of the form

$$\theta \mapsto \exp\Big[-\omega \sum_{i=1}^{n} \big\{ \ell_{\hat{\theta}(Z^{i-1})}(Z_i) - \ell_\theta(Z_i) \big\}\Big]$$

- With "known ω," this is an e-process $\rightsquigarrow$ anytime-validity
- ω must be chosen in applications, but a version of GPC is empirically shown to preserve anytime-validity

[13] Dey, M., and Williams `arXiv:2402.00202`

Conclusion

Generalized Bayes is liberated Bayes. —M

- Normative aspect of Bayesianism is attractive
- But breaking free from fully & correctly specified models creates opportunities for new innovations/applications
- Greater flexibility doesn't come for free, scaling ambiguity is one obstacle $\rightsquigarrow$ learning rate choice
- Lots of different ideas, philosophical considerations
- I've argued that *calibration* be a priority
- Hence, I choose ω specifically targeting calibration

Conclusion, cont.

- Rapid & exciting developments, boundary pushing
- Post-Bayes seminar is timely & valuable — *thanks organizers!*
- Some open questions/problems:
  - prediction — maybe was recently settled?
  - computation, e.g., variational approximations in GPC?
  - more learning rates?
  - reverse engineering loss functions?
  - Gibbs structure learning?
- Other things to think about:
  - intuition is largely based on our understanding of well-specified models, but "All models are wrong"...
  - how much model misspecification can be corrected for?
  - Fraser's "real probabilities" & "hypothetical probabilities"

The end

`https://wordpress-courses2223.wolfware.ncsu.edu/st-790-001-fall-2022/`

- Papers, talks, etc — `www4.stat.ncsu.edu/~rmartin/`
- Questions? — `rgmarti3@ncsu.edu`

Thanks for your interest & attention!